ARTHUR AIMONE

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EDUCATION

Swiss Federal Institute of Technology - EPFL September 2024 - February 2027
MSc. Applied Mathematics, Quantum Science & Computing *Lausanne, CH*

- Relevant courses in Mathematics: Multivariate Statistics, Stochastic Analysis, Stochastic Simulation.
- Relevant courses in Quantum: Quantum Computing, Quantum Information Theory.

King's College London - KCL October 2021 - June 2024
BSc. Mathematics (Grade: 82%) *London, UK*

Specialisation in Probability and Graph Theory, with courses in Quantum Mechanics.

EXPERIENCE

Orange Innovation March 2026 - September 2026
Quantum Computing Research Intern *Rennes, FR*

- Analysis of the impact of quantum computers in Medicine, with an emphasis on drug discovery.
- Methods to implement the fermionic Hamiltonian on a QPU (Trotterisation and orbital rotations).
- Investigated Quantum Subspace Methods for Quantum Chemistry (Krylov and SQD).
- Analysis and implementation of Physics-Informed Learning methods, specifically Neural Quantum States.
- Authored an 80-page technical research report for Orange's Quantum Lab.

Swiss Federal Institute of Technology - EPFL September 2025 - February 2027
Teaching Assistant *Lausanne, CH*

Teaching assistant in a second year course in Probability & Statistics.

PROJECTS

Dynamical Low-Rank Ensemble Kalman Filters for Data Assimilation Fall 2026
MSc. Thesis - Chair of Scientific Computing & Uncertainty Quantification (CSQI) *EPFL*

- Investigating low-rank filtering methods for Data Assimilation in continuous-time.
- Evaluating the impact of covariance localisation on the low-rank filters.
- Implementation of large-scale PDE systems to evaluate the efficacy of the low-rank filtering.

QAOA for the MaxCut Problem on Weighted Random Graphs Fall 2025
Course Project *EPFL*

Solving MaxCut on a random graph with a QAOA circuit in Qiskit and study of the robustness to noise.

Nonparametric Drift Estimation for Stochastic Differential Equations Fall 2025
Semester Project - Chair of Mathematical Statistics (SMAT) *EPFL*

Investigated a nonparametric drift estimation method for SDEs via the Nadaraya-Watson estimator.

TECHNICAL SKILLS

Python (including Qiskit and JAX), LaTeX, GitHub.

INTERESTS

Long distance runner (marathons in Paris & Madrid) and Accordionist.